



Deep marine biosphere

Learn all about the weird and wonderful creatures that live deep beneath the oceanic crust. <http://digit.in/sep20-47>



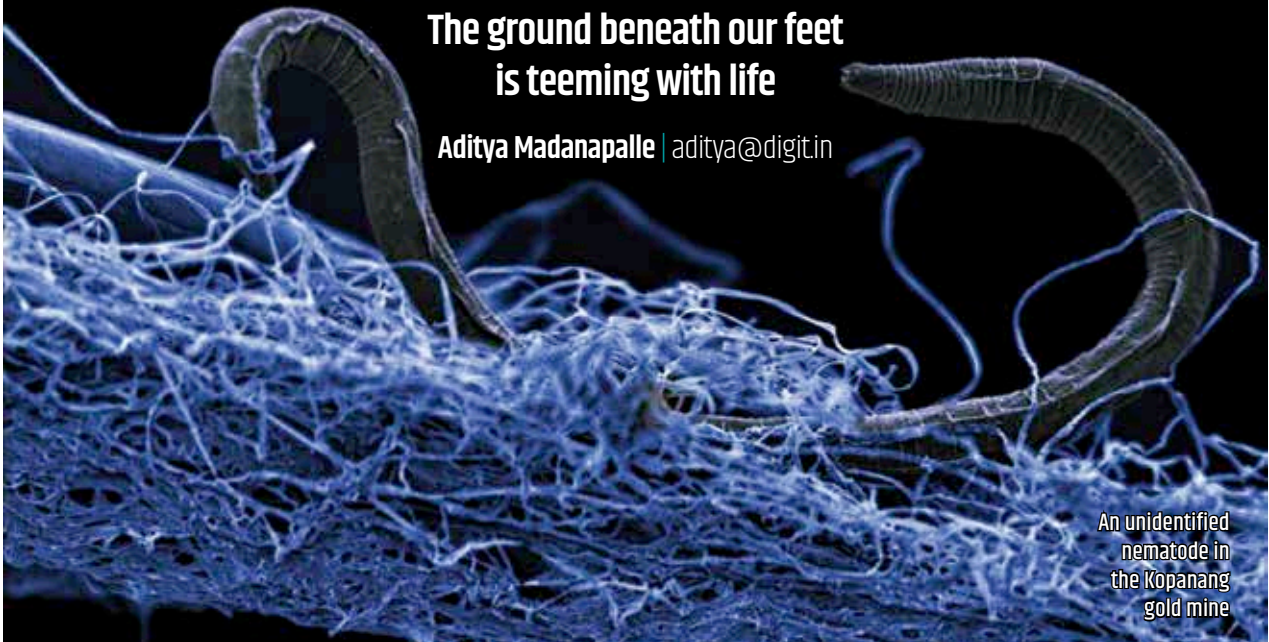
Deep carbon observatory

The deep carbon observatory has been studying the deep biosphere to learn about their impact. <http://digit.in/sep20-48>

Deep Biospheres

The ground beneath our feet
is teeming with life

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An unidentified
nematode in
the Kopanang
gold mine

Credit: Gaetan Borgonie, Extreme Life Isyensis, Belgium

Living things have found a way to survive in almost every place on Earth, from the upper atmosphere to the depths of the ocean. One of the places where we didn't look for life till recently, was deep beneath the ground. These are not the subterranean worms or burrowing mammals, but bacteria, fungi, viruses and even some worms that live deep underground, in conditions of extreme temperatures and pressures, somehow deriving their sustenance from the rocks around them, cut off completely from the sunlight that provides the energy for most of the life on the planet. Still surprisingly, the subterranean biosphere is as full of life as the forests of the Amazon.

The first discovery of these deep biospheres was made by the geologist Edson Bastin, who asked Frank Greer, a microbiologist to investigate the source of hydrogen sulfide and bicarbonates in water extracted from oil fields, from deep beneath the ground. In spring and sea waters these chemicals were known to arise from well understood reactions, involving the metabolism of living microbes. Bastin wanted to find out if the source of the compounds in the waters of oil fields had a similar origin as well, and the experiments proved him right. They managed to culture living bacteria from material extracted in excess of 500 meters beneath the surface, and estimated that the bacteria had to be descendants of creatures buried at least 300 million years ago. Around the same time, Charles Lipman, a

microbiologist from the University of California, Berkeley would take lumps of coal and process them in various ways such as soaking them in water, heating them, and applying pressure on them, before crushing them and using a culture to check for the presence of bacteria. His experiments showed that there were living bacteria in ancient coal deposits as well. He also managed to revive bacteria from sealed bottles of water that were 40 years old. The heating was actually an attempt at sterilising the coal, and baking them at 160-degrees Celsius for 50 hours which actually seemed to increase the number of bacteria that were revived. Scientists knew then that deep beneath the Earth, there were creatures that were living and breathing, and that this life survived, and even thrived